Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Programming & Software Engineering: Python, Java, JavaScript, FastAPI, Flask, REST APIs, Git, GitHub, PyTest, OOPS
Test Automation & Quality: Test Automation, Automated Testing, PyTest, API Testing, Unit Testing, Integration Testing, Test Validation, Edge-Case Testing
AI & Machine Learning: Scikit-learn, XGBoost, Machine Learning, NLP, OCR, RAG, Anomaly Detection, Predictive Modelling, Model Evaluation, Explainable AI
Data & Analytics: SQL, Pandas, NumPy, Data Analysis, Exploratory Data Analysis, Statistical Analysis, Data Cleaning, Feature Engineering, Data Processing
Databases & Data Engineering: MySQL, SQLite, Redis, Relational Databases, Data Warehousing, ETL, Data Pipelines, Structured & Unstructured Data Processing
Visualization & Reporting: Power BI, DAX, Excel, Data Visualization, Dashboard Development, KPI Analysis

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** — ***Aug – Sep, 2025

- Analysed 2 real-world business datasets containing 8,500+ records, applying data preprocessing and AI-assisted analytics to identify trends and generate 15 actionable business insights for data-driven decision-making.
- Validated analytical findings for accuracy and consistency, reviewing outputs to ensure reliable and meaningful insights for business recommendations.
- Established a standardised analytics workflow covering data preprocessing, insight generation, and validation, enabling consistent conversion of raw business data into actionable recommendations.

PROJECTS

***API Rate Limiter Library* |LINK**

- Built a Python rate-limiting library implementing five throttling algorithms, Redis integration, and reusable Flask/FastAPI middleware for API automation and reliable traffic control.
- Designed a modular, thread-safe architecture with pluggable storage backends to support concurrent request handling and reliable API behavior across application environments.
- Developed 30+ PyTest automated tests covering concurrency, edge cases, and algorithm correctness, and packaged the library on PyPI for reusable software development.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built a full-stack Python application using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP to automate receipt processing and reduce processing time by 40%.
- Engineered and validated REST APIs with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export for reliable application workflows.
- Deployed the application on Render and Vercel, implementing end-to-end workflows across AI processing, backend services, database operations, and frontend validation.

***Docwise* |LINK**

- Engineered an end-to-end RAG-based AI document pipeline using Google Gemini, NLP, TF-IDF retrieval, and intelligent chunking, achieving 95%+ query accuracy on PDFs exceeding 200 pages.
- Developed automated document-processing workflows covering extraction, chunking, retrieval, contextual ranking, and LLM-based response generation.
- Improved AI output reliability and validation through retrieval tuning, prompt engineering, and hallucination-aware checks, increasing contextual relevance and answer consistency.

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered a Python analytics pipeline processing 50K+ transactions, combining statistical analysis, duplicate detection, vendor-velocity checks, HHI analysis, and Isolation Forest for automated anomaly detection.
- Designed a SQLite star-schema warehouse and composite 0–100 risk scoring pipeline, identifying 357 high/critical-risk transactions with automated analysis completed in under 7 seconds.
- Built a 5-page Power BI audit dashboard with 20+ DAX measures and automated pipeline tests to validate data quality, anomaly detection, and reporting reliability.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026